

LinkedIn Job Trend Analysis

Project 14 | Domain: Data Analytics & Web Scraping | Tools: Python, BeautifulSoup, Pandas, Matplotlib, Seaborn, Excel

1. Introduction

The job market for technology and data roles evolves rapidly, with in-demand skills shifting based on industry needs, geography, and emerging technologies. Understanding these trends is critical for job seekers, hiring managers, and academic institutions alike. This project addresses this need by programmatically scraping LinkedIn's public job feed across multiple roles and cities in India, extracting structured skill data, and performing comprehensive trend analysis. The result is a data-driven picture of which skills are most demanded, how skill requirements differ across cities, and which roles offer the best match for a given candidate's profile.

2. Abstract

This project implements an end-to-end LinkedIn job market intelligence pipeline. Using Python's BeautifulSoup library, job postings were scraped from LinkedIn's public unauthenticated feed with User-Agent rotation and randomized request throttling to simulate human-like browsing behavior. Postings were collected across four tech roles — Data Analyst, Data Scientist, ML Engineer, and Software Engineer — across major Indian cities including Bengaluru, Mumbai, Delhi NCR, Hyderabad, and Pune. A curated skill keyword library of 20 in-demand technologies (Python, SQL, Tableau, Power BI, AWS, Spark, Docker, etc.) was matched against job descriptions. Cross-tabulation matrices were generated to map skill frequency by role and by city. A recommendation engine was built to match a candidate's skill set against open postings and recommend best-fit roles and high-hiring cities. All results were exported to Excel with heatmap visuals.

3. Tools Used

- **Python 3.x** — Core scripting language for scraping, data processing, and analysis
- **BeautifulSoup4 & Requests** — Web scraping of LinkedIn public job feed with HTML parsing
- **Pandas & NumPy** — Data cleaning, skill extraction, cross-tabulation, and matrix operations
- **Matplotlib & Seaborn** — Heatmaps, bar charts, and skill frequency visualizations
- **Regex (re)** — Pattern matching to extract job titles, locations, and skill tags from raw HTML
- **Excel (openpyxl)** — Final structured output of skill matrices, role summaries, and recommendations
- **User-Agent Rotation** — Randomized browser headers to prevent bot detection during scraping

4. Steps Involved in Building the Project

Step 1. Environment Setup & Skill Library Definition

Configured professional Seaborn plotting parameters. Defined a curated library of 20 high-demand tech skills: Python, SQL, R, Tableau, Power BI, Excel, AWS, Azure, Spark, Hadoop, Scikit-learn, TensorFlow, PyTorch, Snowflake, Docker, Kubernetes, Airflow, Git, Java, and Scala.

Step 2. LinkedIn Public Feed Scraping

Built a scraper function targeting LinkedIn's public job board URL with role and location as URL parameters. Implemented User-Agent rotation across 5 browser profiles and randomized sleep intervals (2–5 seconds) between requests to mimic human browsing. Scraped 3 pages (75 postings) per role-city combination.

Step 3. HTML Parsing & Metadata Extraction

Used BeautifulSoup to parse HTML responses. Extracted job title, company name, location, and full job description text. Handled missing or malformed records with try-except guards, logging failures without stopping the pipeline.

Step 4. Skill Keyword Extraction

Applied a `clean_and_extract_skills()` function that scanned raw job description strings for matches against the 20-keyword library using case-insensitive comparison. Stored matched skills as a comma-separated string per posting.

Step 5. Cross-Tabulation Matrix Generation

Exploded comma-separated skill lists into individual rows. Generated two cross-tabulation matrices: (i) Skill × Role matrix — showing skill frequency normalized by role; and (ii) Skill × City matrix — showing skill density normalized by city. Values expressed as percentages.

Step 6. Visualization — Heatmaps & Bar Charts

Plotted Skill × Role and Skill × City heatmaps using Seaborn with annotated values. Generated bar charts for top 10 most in-demand skills overall. Saved all figures as high-resolution PNG outputs.

Step 7. Recommendation Engine

Built a `labor_market_recommendation_engine()` function accepting a user's skill list. Computed skill overlap scores against all scraped postings to rank best-fit roles and highest-hiring cities for the given profile. Outputs top 3 role matches and top 3 city matches with percentage alignment scores.

Step 8. Excel Export

Compiled all outputs — scraped data, skill matrices, role summaries, and personalized recommendations — into a structured Excel workbook with separate sheets for each analysis layer.

5. Conclusion

The LinkedIn Job Trend Analysis project delivered a fully automated, data-driven view of the Indian technology job market. The scraping pipeline successfully collected and structured real-world job postings across four roles and five major cities. Key findings indicated that Python, SQL, and Power BI consistently ranked as the top three demanded skills across all Data Analyst and Data Scientist roles, while Docker, Kubernetes, and AWS dominated Software Engineer and ML Engineer postings. Geographically, Bengaluru showed the highest density of ML-specific roles, while Pune and Mumbai exhibited stronger demand for BI and reporting tools. The recommendation engine provided personalized, data-backed guidance for candidates to optimize their job search strategy. This project demonstrated practical skills in web scraping, NLP-style text parsing, matrix analysis, and translating raw job market data into actionable career intelligence.